Student–Faculty Co-Creation of Open Educational Resources for Learning Applied Statistics with Open Source Software Tools

Nurlana Alili [1], Xi (Bryan) Su [1], Nishan Mudalige [1], Masoud Ataei [1]

[1] Department of Mathematical and Computational Sciences, University of Toronto Mississauga, Mississauga, ON L5L 1C6, Canada

Background and Introduction

Student–faculty partnerships enhance learning by increasing engagement, motivation, and authenticity, yet such collaborations are rarely documented in statistics. To address this gap, students enrolled in *STA378: Statistics Research Project* co-created an e-book for *STA258: Statistics with Applied Probability* at UTM. The e-book integrates narrative text, simulations, visualizations, and interactive apps to modernize course content and reduce redundancy. Our project offers a practical model for student–faculty partnerships and forms part of a long-term redevelopment project for STA258.

Results and Significance

By leveraging open source software to build an accessible, interactive resource, we created a sustainable model others can adapt. Our experience demonstrates that participation in curriculum development can produce engaging materials while empowering students as active contributors. We also believe this is one of the few projects involving student–faculty co-creation of materials in statistics, and potentially one of the first projects of its kind in a Canadian context.

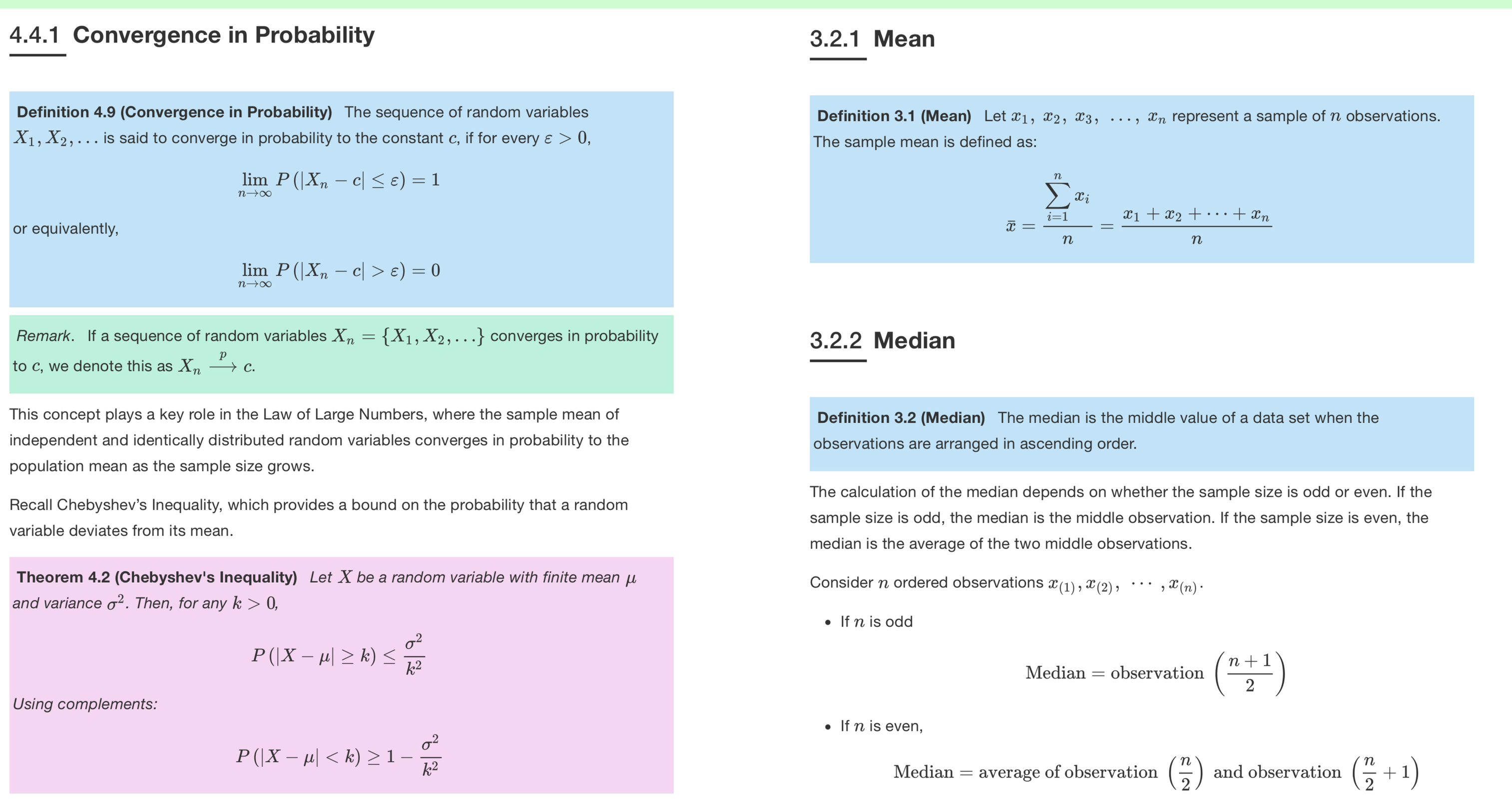


Figure 2: Examples of text content, including math statements and customized environments.

Methodology and Implementation

The project was completed using open source tools. Content was written in Markdown and rendered using the Bookdown framework. Source files are managed on GitHub with the e-book hosted on GitHub Pages. Figures are rendered within code blocks and math statements are written in LaTeX within Markdown. Shiny applications are hosted on shinyapps.io and embedded directly in the e-book using HTML. Custom styling was implemented in CSS with formatting, metadata, and output options controlled using YAML.

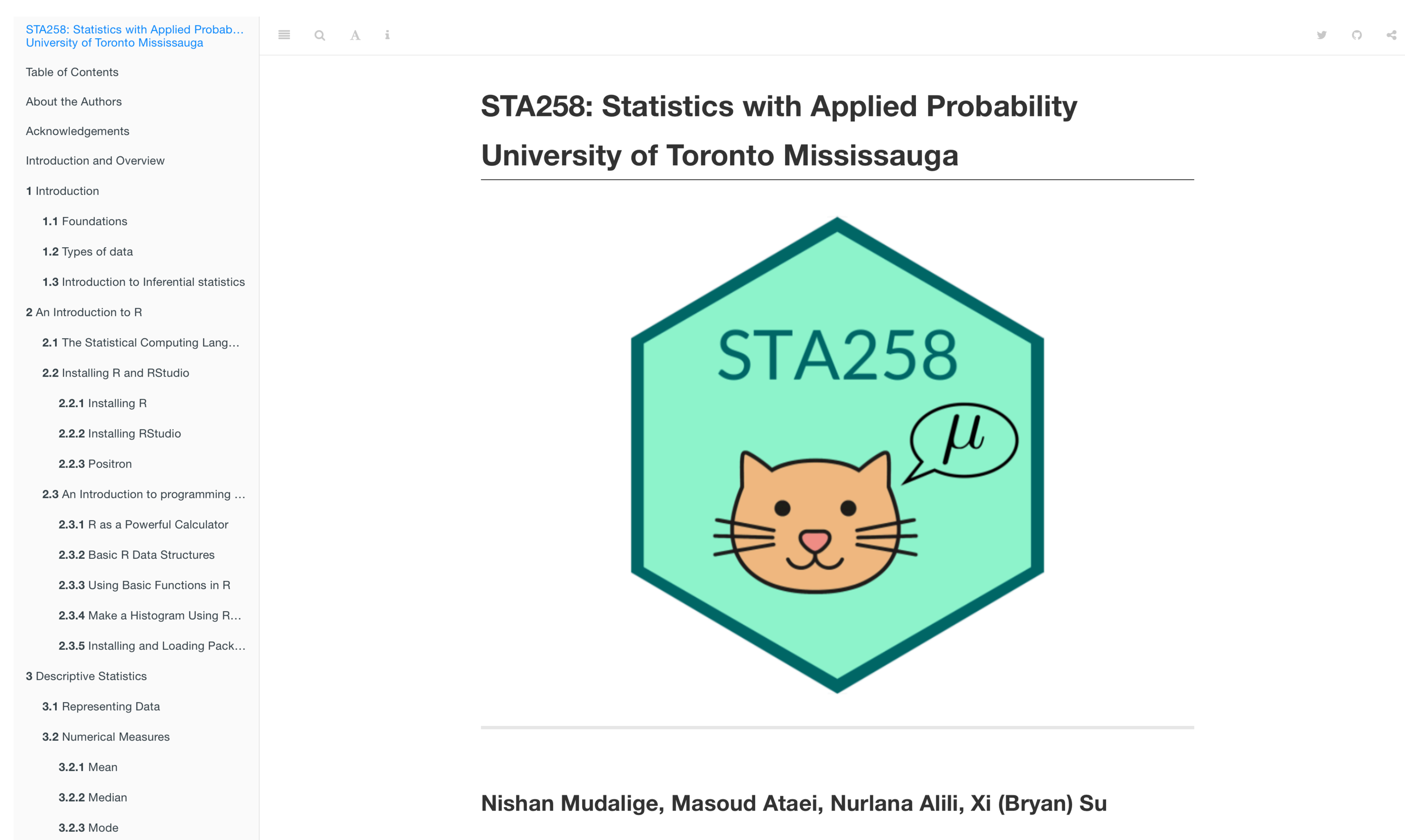


Figure 1: Landing page of the e-book. The hex sticker is an `rgl` object. The table of contents is on the left pane. The menu bar at the top contain accessibility options, has a search feature and can hide the table of contents (Also see Figure 6).

Available at *https://nishanmudalige.github.io/STA258/*

Advantages of the Frameworks Used

- Open source: Freely available to the broader community; actively maintained.
- Self-contained: Text written in Markdown; figures rendered in code blocks.
- Math-friendly: Math statements typeset in LaTeX are rendered with MathJax.
- Interactive: Shiny apps embedded directly with simple HTML.
- Customizable: Fonts, colors, and layouts modified through YAML and CSS.
- Accessible: Dark mode, high-contrast mode, and quick search are embedded.
- Integration: Plots, figures, and apps are rendered directly from R code blocks.
- Version-control friendly: Maintained on GitHub, reproducible by design.
- Affordable: E-book hosted on GitHub Pages (free); apps on shinyapps.io.
- Future-proof: Sustained with open source tools and community contributions via forking and pull requests.

Languages	CSS, HTML, JavaScript, LaTeX, Markdown, R, YAML
Frameworks	Bookdown, R Markdown
R packages	`bookdown, dplyr, ggplot2, knitr, plotly, rgl, Shiny`
Version	Git (with GitHub)
Deployment	GitHub Pages, shinyapps.io

Table 1: Summary of languages, packages, frameworks, and other tools used do develop the e-book.

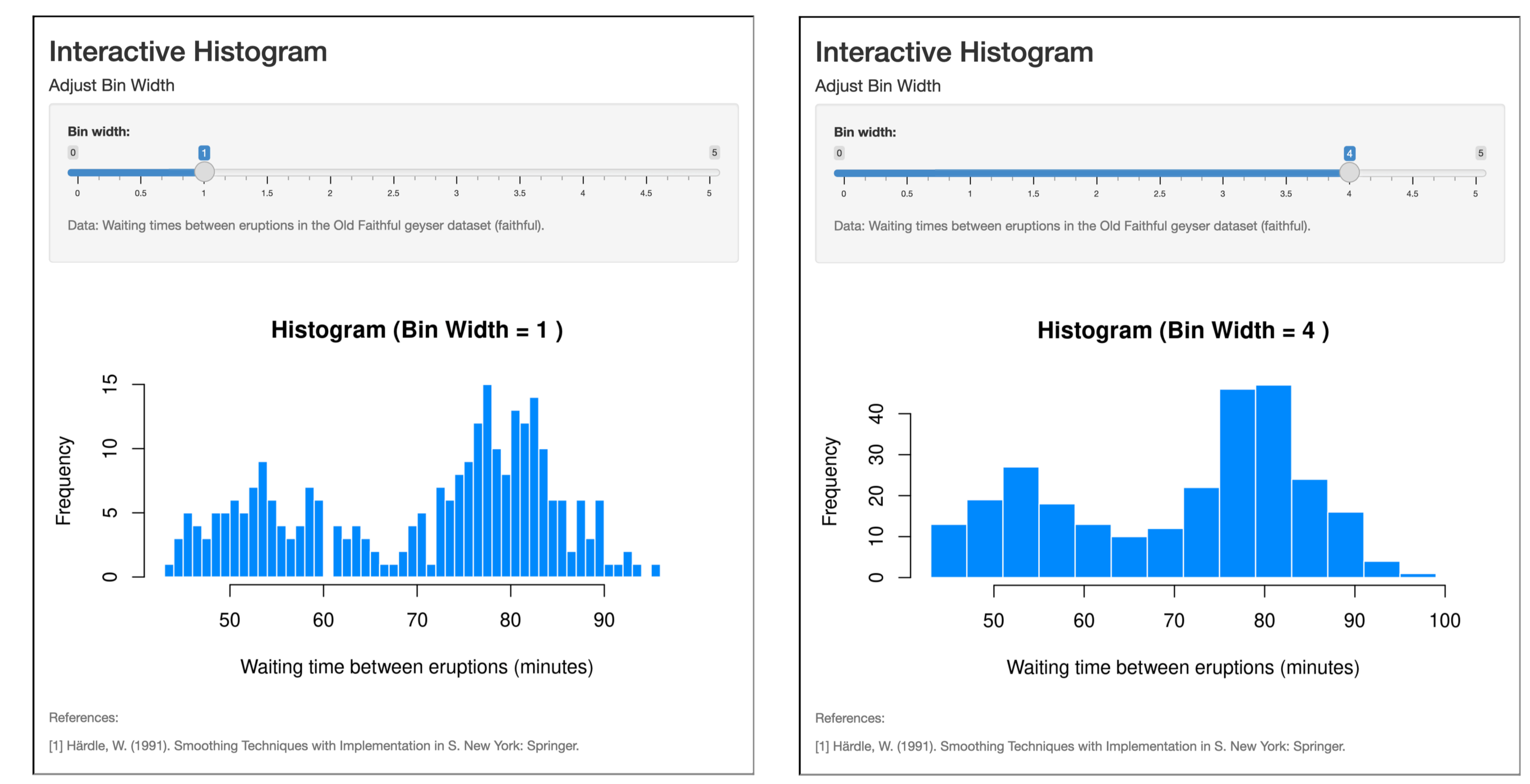


Figure 3: Example of an interactive application made with `shiny` to demonstrate the effect of bin widths on histograms.

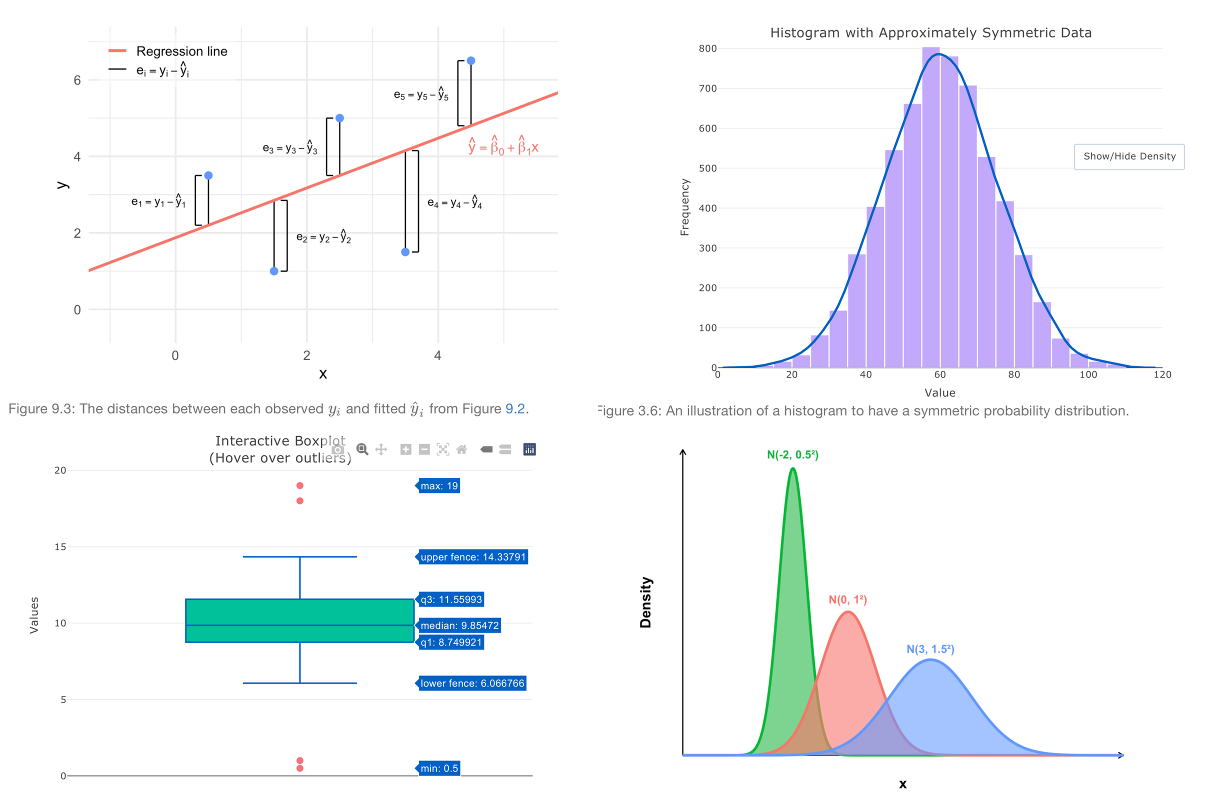


Figure 4: Examples of static and interactive plots created with `ggplot` and `plotly`.

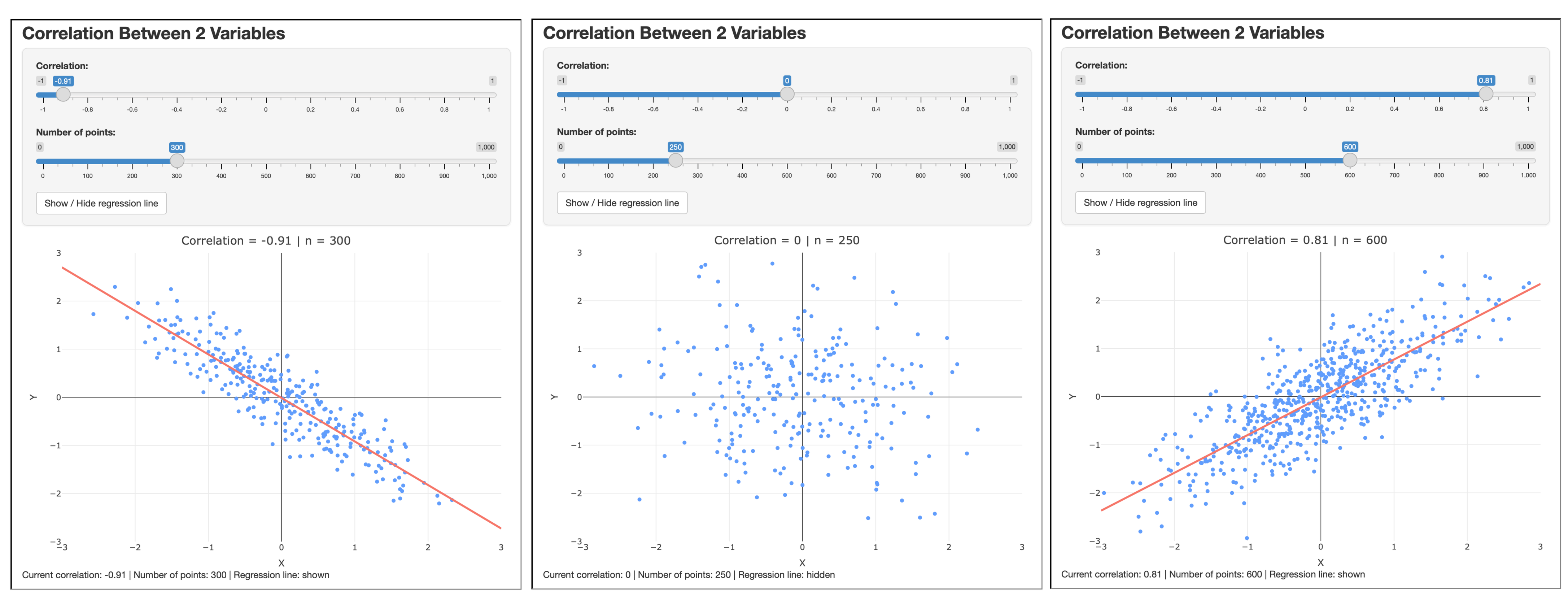


Figure 5: Example of an interactive application made with `shiny` to demonstrate correlation.

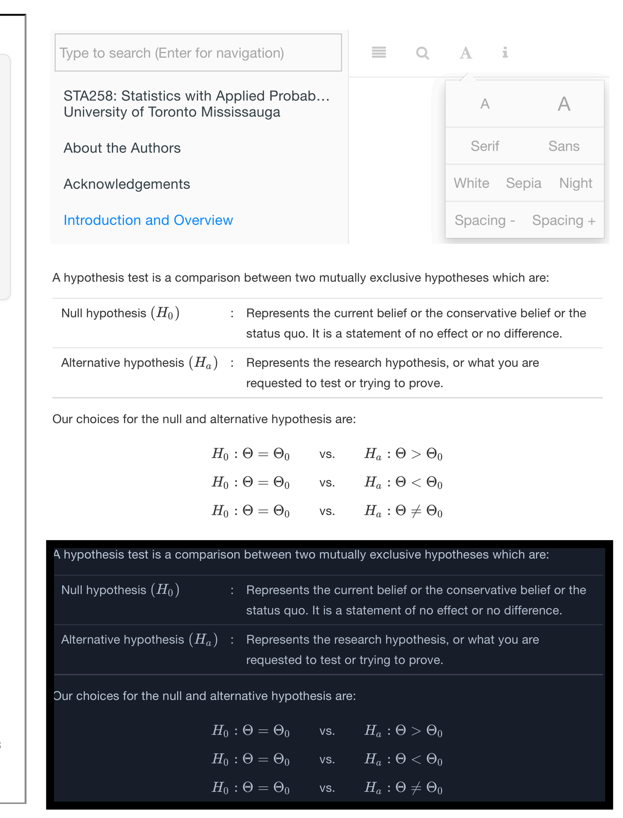


Figure 6: Accessibility and search features.

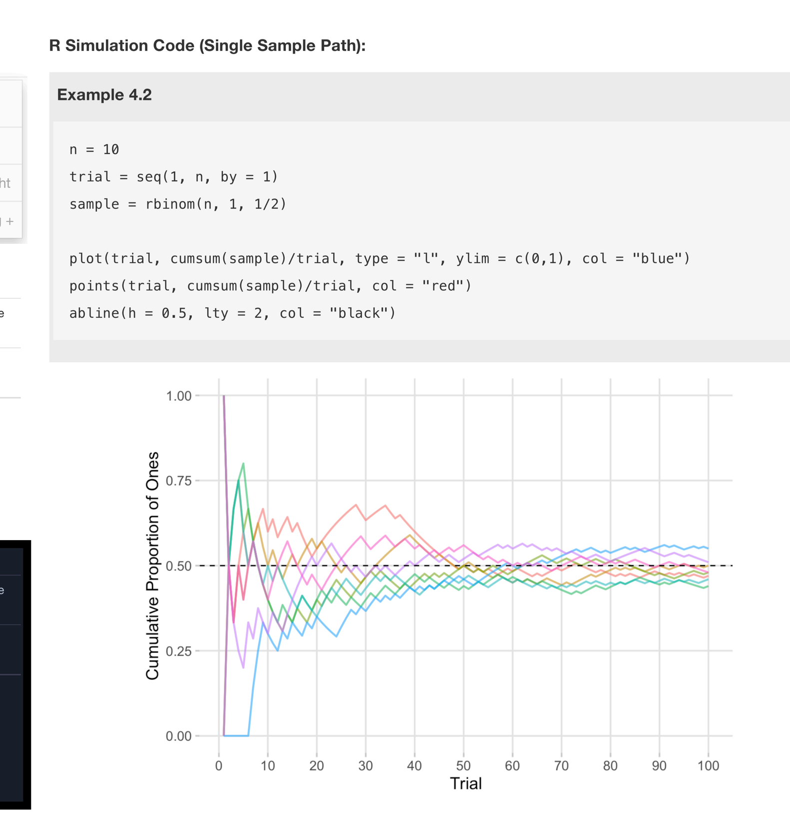
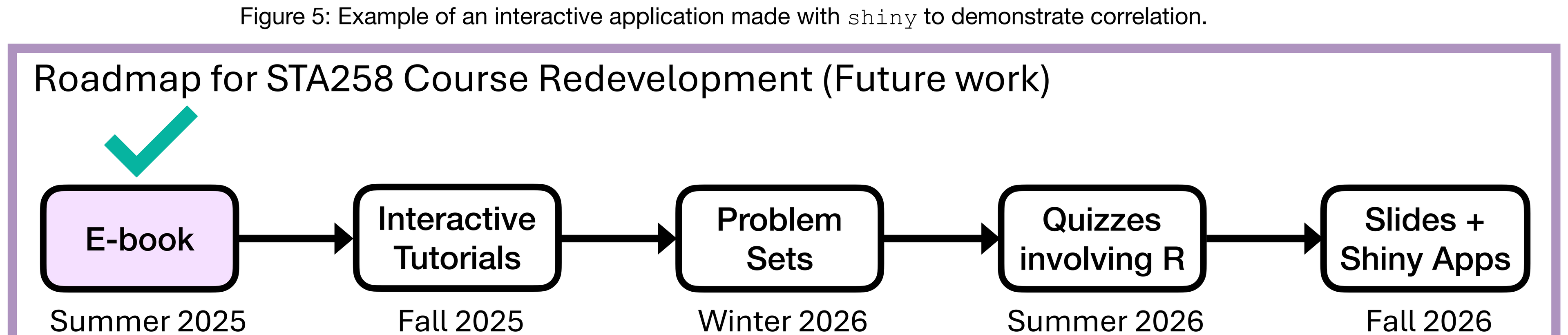


Figure 7: Examples of a code block and exercise with R.

Conclusions

Through this project, students learned the value of working across technical, design, and communication challenges in a collaborative setting with faculty and external partners. In addition to learning new skills, the experience introduced students to project management, and enhanced their creativity and problem-solving abilities beyond the classroom. The e-book serves as a course resource and a model for student involvement in shaping academic materials.



Made with

bookdown, rmarkdown, knitr, ggplot2, plotly, Shiny, dplyr, RStudio

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